GRADE 7 ACCELERATED · UNIT 6 · LESSON 4

Pythagorean Theorem in Real-World Problems

Question Bank - Additional 6.04 **ANSWER KEY**

Name
Date
Period

Show your work. Circle your final answer.

For questions 1 through 7, sketch the right triangle first. Round to the nearest tenth when the answer is not a whole number.

1. A window is 8 ft wide and 6 ft tall. How long is its diagonal?

10 ft

64 + 36 = 100.

2. An 11-ft ladder leans on a wall with its foot 4 ft from the wall. How high does it reach?

About 10.2 ft

121 − 16 = 105; √105 = 10.246…, so 10.2.

3. A ramp rises 3 ft over a run of 10 ft. How long is the ramp?

About 10.4 ft

9 + 100 = 109; √109 ≈ 10.44.

4. A garden bed is 3 m by 4 m. How much shorter is the walk straight across, corner to corner, than the walk along two sides?

2 m

Along two sides 7 m, across 5 m.

5. Will a 15-in rod lie flat in a box 9 inches by 11 inches, corner to corner?

No — the diagonal is only about 14.2 in.

81 + 121 = 202; √202 ≈ 14.21.

6. A kite's 17-m string is held at a spot 8 m from the point straight below the kite. How high is the kite?

15 m

289 − 64 = 225; the string is c.

7. A wire runs from the top of a 9-m pole to a stake in the ground 12 m from the pole's base. How long is the wire?

15 m

81 + 144 = 225.

Choose one.

8. A 10-ft ladder reaches 8 ft up a wall. How far from the wall is its foot?

- ● **A. 6 ft**
- ○ B. 2 ft
- ○ C. 36 ft
- ○ D. 12.8 ft

A

$100 - 64 = 36 = 6^2$.

9. A wire runs from the top of a pole to a stake in the ground. Which length is the hypotenuse?

- ○ A. the pole
- ● **B. the wire**
- ○ C. the ground from the pole to the stake
- ○ D. none of them

B

The pole stands square on the ground; across from that corner is the wire.